STATISTICS · SAMPLING AND EXPERIMENTATION

Random Assignment Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Random Assignment Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Students choose whether to use a study app, and stronger students tend to choose it. If the app does nothing, what might the experiment show? Why?

2. Does random assignment guarantee that the two groups have the same mean GPA?

2 Explore use the tool

1. True effect 0, Students choose, 200+ repetitions: record the center and SD of the differences. What do you conclude?

2. Random assignment, true effect 0, 200+ repetitions: record the center and SD.

3. Random assignment: assign once at a time. How many tries until the app group's mean GPA is at least 0.3 higher? What does that show?

4. Same true effect: compare random assignment with randomized pairs (200+ each). Record both SDs and explain the difference.

3 Record use the tool and fill in the table

Assignment methods (true effect +4)

200 or more repetitions each. Values vary a little; the key gives long-run values.

Assignment	Center of differences	SD	Unbiased?
Random assignment			
Randomized pairs			
Students choose			

Randomization test: how often is p < 0.05?

Run 10 experiments (1,000 re-randomizations each) for each true effect.

True effect	0	+4	+8	+12
Experiments with p < 0.05 (out of 10)				
Long-run share				

4 Explain what did you discover?

1. When students choose their group, GPA is _____ with the treatment, so we can't tell which one caused the difference.
2. Random assignment makes the groups similar _____, not in every single experiment.
3. Randomizing within matched pairs _____ the variability of the estimates.
4. A randomization-test p-value is the share of shuffles with a difference _____ the observed one, assuming _____.
5. A randomization test for the app gives a p-value of 0.31. Explain what this number means in context, and what you can conclude.

5 Apply a new situation, no tool

1. Students who chose to attend tutoring had higher grades than students who didn't. Why can't we conclude that tutoring caused the higher grades? Describe an experiment that could.

2. Identify the design: 30 pairs of twins; in each pair a coin flip decides which twin gets the new program.

3. In a randomization test with 500 shuffles, 9 differences were at least as large as the observed one. Find the p-value and state a conclusion at $\alpha = 0.05$.